REPORT

CHURN SCOPE:

AI-POWERED CUSTOMER RETENTION AND CHURN PREDICTION SYSTEM

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INTRODUCTION

1.1 SYNOPSIS

Customer churn has become one of the most significant challenges faced by telecommunication companies in today's highly competitive market. Acquiring new customers requires substantial investment in marketing, promotions, and customer acquisition strategies, while retaining existing customers is comparatively more cost-effective. Therefore, understanding customer behaviour and identifying potential churners before they leave has become a crucial business objective.

This project, titled "Churn Scope: AI-Powered Customer Retention and Churn Prediction System", focuses on analysing customer data and predicting whether a customer is likely to discontinue the service. The project utilizes the Telco Customer Churn dataset, which contains demographic information, service details, billing information, and customer subscription records.

The project workflow includes data preprocessing, exploratory data analysis, correlation analysis, feature engineering, machine learning model development, model evaluation, and retention recommendation generation. Important customer attributes such as contract type, tenure, monthly charges, and internet service are analysed to identify factors influencing customer churn.

Machine learning algorithms including Logistic Regression, Decision Tree, and Random Forest are implemented and compared. The model with the best performance is selected as the final prediction model. In addition, a retention recommendation engine is developed to suggest suitable actions for customers with high churn probability.

The proposed system helps organizations proactively identify at-risk customers and implement effective retention strategies. By combining business analysis with machine learning techniques, the project contributes to improving customer satisfaction, reducing revenue loss, and enhancing long-term customer relationships.

1.2 ABOUT THE PROJECT

Customer retention is a key performance indicator for telecommunication companies. High churn rates can directly impact profitability and market share. Traditional customer management approaches often fail to identify customers who are likely to leave before churn actually occurs.

The objective of this project is to develop an intelligent customer retention system using machine learning techniques. The system analyses customer characteristics and predicts the likelihood of churn. Based on the prediction results, customers are categorized according to risk levels, allowing businesses to take preventive actions.

The project utilizes a real-world customer churn dataset containing information such as customer demographics, service subscriptions, billing methods, contract details, and churn status. Through data preprocessing and exploratory data analysis, meaningful patterns are extracted from the dataset.

Several machine learning algorithms are trained and evaluated to determine the most suitable model for churn prediction. The project also incorporates feature importance analysis to identify the most influential variables affecting customer retention.

The final outcome is an AI-powered decision support system capable of predicting customer churn and recommending appropriate retention strategies.

1.3 INTRODUCTION

The rapid growth of the telecommunications industry has increased competition among service providers. Customers now have multiple options available and can easily switch to competitors if their expectations are not met. As a result, customer churn has become a major concern for businesses.

Customer churn refers to the phenomenon where customers discontinue a company's products or services and move to an alternative provider. Understanding why customers leave is essential for developing effective retention strategies.

Data Science and Machine Learning provide powerful techniques for analysing customer behaviour and predicting future actions. By leveraging historical customer data, organizations can identify patterns associated with churn and intervene before customers decide to leave.

The customer churn prediction process typically involves collecting customer information, preprocessing data, analysing customer behaviour, building predictive models, and evaluating model performance. These predictions can help organizations optimize customer retention campaigns and improve business performance.

This project applies data science methodologies to develop a practical churn prediction system. Various customer-related factors are analysed to determine their impact on churn. Machine learning algorithms are used to classify customers as likely to churn or remain with the company.

The project demonstrates how predictive analytics can be used to transform raw customer data into actionable business insights.

1.4 DATA OVERVIEW

The dataset used in this project is the Telco Customer Churn Dataset, which contains information about customers of a telecommunications company. It includes customer demographics, service details, billing information, and churn status. The dataset contains around 7,000 customer records and is widely used for analysing customer behaviour and predicting churn.

Important Attributes

Attribute	Description
customerID	Unique customer identifier
gender	Customer gender
SeniorCitizen	Indicates whether customer is a senior citizen
Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed with company
PhoneService	Availability of phone service
InternetService	Type of internet service
OnlineSecurity	Online security subscription
TechSupport	Technical support subscription
Contract	Contract type
PaperlessBilling	Billing method
PaymentMethod	Customer payment method
MonthlyCharges	Monthly service charges

TotalCharges	Total amount charged
Churn	Target variable indicating customer churn

1.5 OBJECTIVES OF THE PROJECT

The primary objectives of this project are:

1. To analyse customer churn behaviour using the Telco Customer Churn dataset.
2. To identify important factors influencing customer churn.
3. To perform exploratory data analysis and discover meaningful business insights.
4. To develop machine learning models for churn prediction.
5. To compare multiple classification algorithms and select the best-performing model.
6. To evaluate model performance using suitable metrics.
7. To identify the most important features contributing to churn.
8. To generate churn risk scores for customers.
9. To develop a retention recommendation engine.
10. To support business decision-making through predictive analytics.

1.6 SCOPE OF THE PROJECT

The scope of this project includes customer churn analysis, predictive modelling, and retention strategy development within the telecommunications domain.

The project focuses on:

- Customer behaviour analysis.
- Churn prediction using machine learning.
- Business insight generation through exploratory data analysis.
- Risk-based customer segmentation.
- Retention recommendation generation.

The developed system can assist telecommunication companies in identifying customers likely to leave and taking preventive measures to improve customer retention.

Future enhancements may include real-time churn prediction, cloud deployment, dashboard integration, and advanced deep learning techniques for improved prediction accuracy.

2. DATA PREPROCESSING

Data preprocessing is one of the most important stages in any data science project. Raw datasets often contain inconsistencies, missing values, duplicate records, and categorical variables that cannot be directly processed by machine learning algorithms. Therefore, preprocessing is required to transform the raw data into a clean and structured format suitable for analysis and model development.

The Telco Customer Churn dataset contains customer demographic information, service subscriptions, billing details, and churn status. Several preprocessing techniques were applied to improve data quality and prepare the dataset for machine learning modelling.

2.1 IMPORTING LIBRARIES

- **Pandas**
  Pandas was used for loading, cleaning, and manipulating the dataset. It provides efficient data structures for handling tabular data.
- **NumPy**
  NumPy was used for numerical operations and mathematical computations.
- **Matplotlib**
  Matplotlib was utilized to generate charts and visualizations required for exploratory data analysis.
- **Seaborn**
  Seaborn was used to create advanced statistical visualizations such as heatmaps and boxplots.
- **Scikit-Learn**

Scikit-Learn libraries were used for preprocessing, model training, prediction, and evaluation.

```
# ==========================================
# Import Libraries
# ==========================================

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

2.2 DATA LOADING

The Telco Customer Churn dataset was loaded using the Pandas library. The dataset contains customer records along with demographic information, subscription details, billing information, and churn status.

After loading the dataset, the structure of the data was examined using functions such as:

- head()
- info()
- describe()
- shape()

These functions helped identify the number of rows, columns, data types, and statistical properties of the dataset.

```python
# ==========================================
# Load Dataset
# ==========================================

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
```

2.3 DATA CLEANING

Data cleaning was performed to improve data quality and eliminate inconsistencies.

Duplicate Records

The dataset was checked for duplicate records. Duplicate entries can introduce bias into machine learning models and affect analytical results. Any duplicate records identified were removed.

Missing Values

The dataset was examined for missing values. Missing entries can negatively impact predictive performance if not handled properly.

The TotalCharges column contained blank values which were converted into missing values and removed during preprocessing.

The dataset was verified after cleaning to ensure consistency and completeness.

2.4 DATA CONVERSION

Certain variables required conversion into appropriate formats.

Total Charges

The TotalCharges attribute was initially stored as a text field. It was converted into a numerical format to facilitate mathematical calculations and machine learning operations.

Data Type Verification

All dataset attributes were reviewed to ensure they possessed suitable data types for analysis and modelling.

The conversion process improved computational efficiency and prevented errors during model development.

2.5 FEATURE ENCODING

Machine learning algorithms require numerical input data. Therefore, categorical variables were converted into numerical representations using Label Encoding.

The following categorical features were encoded:

- Gender
- Partner
- Dependents
- Phone Service
- Internet Service
- Contract
- Payment Method
- Churn

```
# ==========================================
# Feature Encoding
# ==========================================

encoder = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = encoder.fit_transform(df[col])
```

Each unique category was assigned a numerical value.

Feature encoding allowed machine learning algorithms to process categorical information effectively.

2.6 FEATURE SELECTION

Feature selection was performed to identify relevant variables contributing to churn prediction.

The target variable selected for the project was:

Churn

The remaining attributes were considered independent variables.

Some identifiers such as customerID were excluded because they do not contribute meaningful information for prediction.

The selected features included:

- Tenure
- Contract
- Internet Service
- Monthly Charges
- Total Charges
- Payment Method
- Online Security
- Tech Support

These variables were expected to influence customer retention behaviour.

```
# ==========================================
# Feature Selection
# ==========================================

X = df.drop(
    ['customerID','Churn'],
    axis=1
)

y = df['Churn']
```

2.7 DATA SPLITTING

The dataset was divided into training and testing subsets.

Training Set

The training dataset was used to teach machine learning models underlying customer behaviour patterns.

Testing Set

The testing dataset was used to evaluate model performance on unseen data.

A standard split ratio of:

70% Training Data

30% Testing Data

was applied.

This approach ensured unbiased evaluation of model performance.

```
# ==========================================
# Train Test Split
# ==========================================

X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_state=4
```

2.8 NORMALIZATION AND SCALING

Feature scaling was applied to numerical variables to ensure consistent value ranges.

The following numerical features were scaled:

- Tenure
- Monthly Charges
- Total Charges

Scaling improves machine learning performance by preventing variables with larger magnitudes from dominating model training.

Standard scaling was used to normalize numerical attributes.

EXPLORATORY DATA ANALYSIS

3. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is a crucial stage in the data science workflow. It helps in understanding the characteristics of the dataset, identifying hidden patterns, discovering relationships among variables, and generating meaningful business insights.

In this project, EDA was performed to analyze customer behaviour and identify factors that significantly influence customer churn. Various visualizations such as bar charts, boxplots, and heatmaps were used to examine the relationship between customer attributes and churn status.

The analysis focused on contract type, tenure, monthly charges, internet service, and overall customer churn trends.

3.1 OVERALL CHURN RATE ANALYSIS

Customer churn represents the percentage of customers who discontinue the company's services.

The Telco Customer Churn dataset contains approximately 7043 customer records, out of which around 1869 customers have churned.

The overall churn rate is approximately 26.5%, indicating that nearly one out of every four customers leaves the company.

This observation highlights the importance of customer retention strategies. A high churn rate can significantly affect revenue generation and business growth.

Key Observation

- Approximately 26.5% of customers churned.
- Around 73.5% of customers remained with the company.
- Customer retention remains a major business challenge.

Output Reference: Churn Distribution Chart

```
Churn                  No        Yes
Contract
Month-to-month  57.290323  42.709677
One year        88.722826  11.277174
Two year        97.151335   2.848665
```

3.2 CONTRACT TYPE ANALYSIS

Contract type is one of the most influential factors affecting customer churn.

The dataset contains three contract categories:

- Month-to-Month
- One Year
- Two Year

The analysis revealed that customers with month-to-month contracts exhibit significantly higher churn rates compared to customers with long-term contracts.

Customers under one-year and two-year contracts demonstrated greater loyalty and lower churn probabilities.

Key Findings

Contract Type	**Churn Trend**
Month-to-Month	Highest
One Year	Moderate
Two Year	Lowest

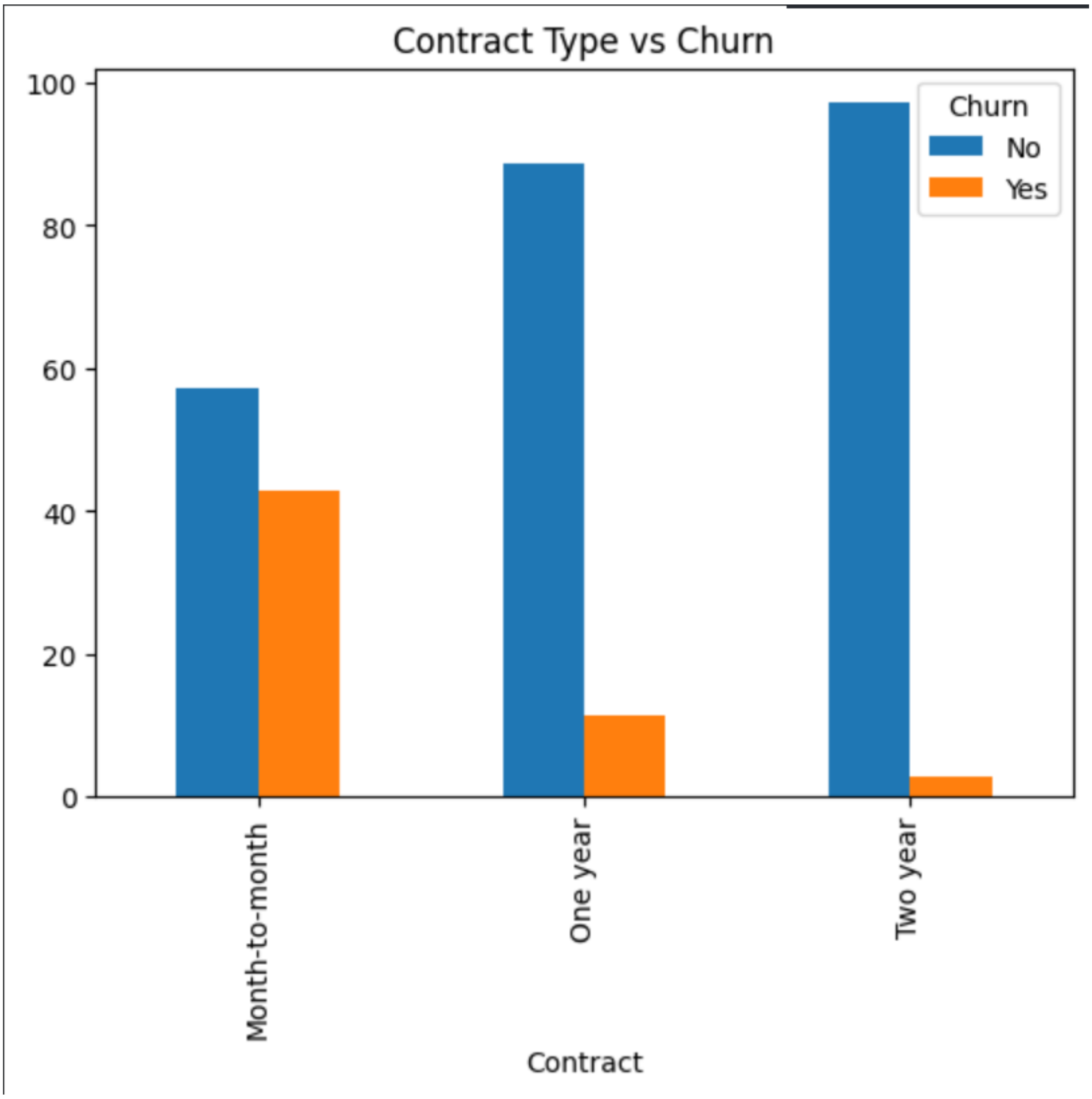
Business Interpretation

Long-term contracts encourage customer commitment and reduce the likelihood of switching to competitors.

Key Observation

Contract type emerged as one of the strongest predictors of customer churn.

Output Reference: Contract Type vs Churn Bar Chart



3.3 TENURE ANALYSIS

Tenure represents the duration for which a customer has remained with the company.

The analysis showed that customers with shorter tenure are more likely to churn, while customers with longer tenure exhibit greater loyalty.

The boxplot visualization clearly demonstrates that churned customers generally have lower tenure values compared to retained customers.

Key Findings

- New customers have the highest churn rates.
- Long-term customers are less likely to leave.
- Customer loyalty increases with tenure.

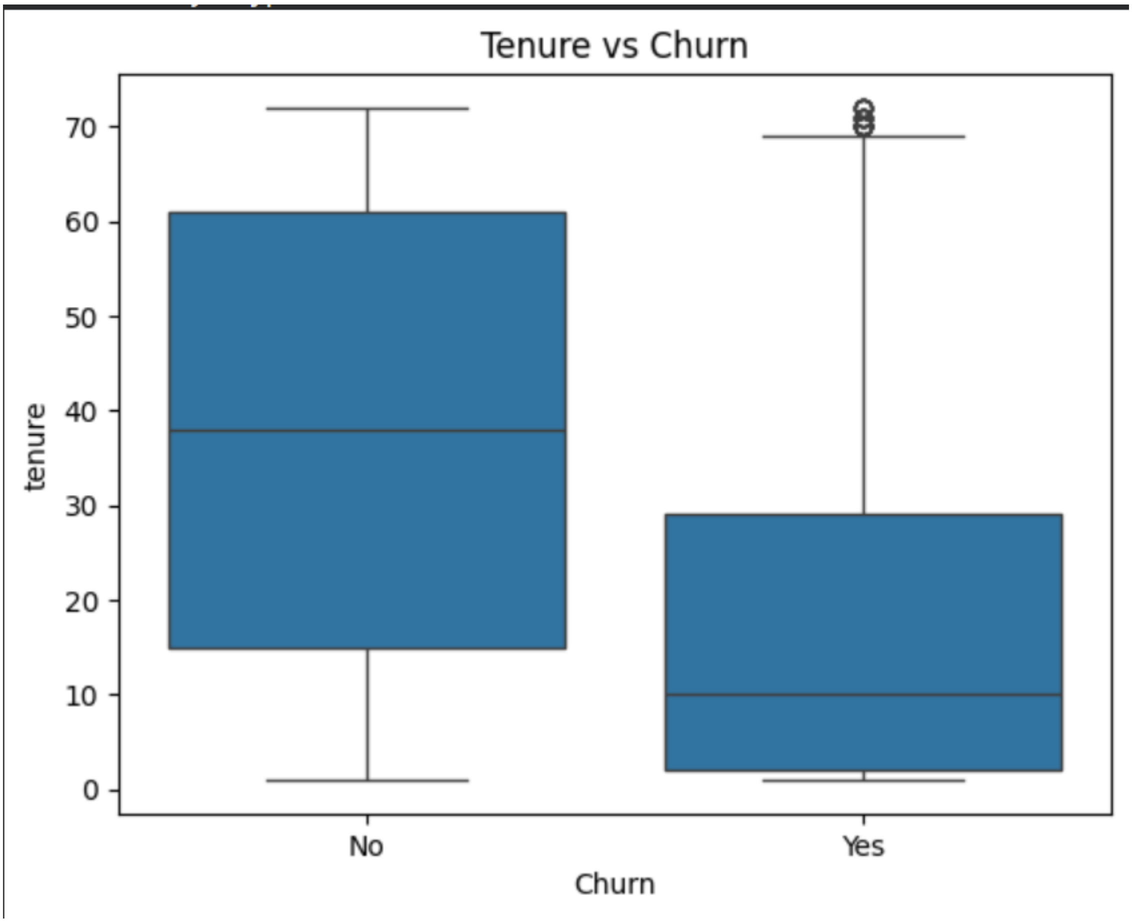
Business Interpretation

The initial months of customer engagement are critical for retention. Businesses should focus on onboarding and customer satisfaction during the early stages of subscription.

Key Observation

Tenure has a strong negative relationship with churn.

Output Reference: Tenure vs Churn Boxplot



3.4 MONTHLY CHARGES ANALYSIS

Monthly charges represent the amount paid by customers each month for services.

The analysis revealed that customers with higher monthly charges tend to churn more frequently than customers with lower monthly charges.

This suggests that pricing plays a significant role in customer retention.

Key Findings

- Higher monthly charges increase churn probability.
- Customers paying lower monthly charges are more likely to remain subscribed.

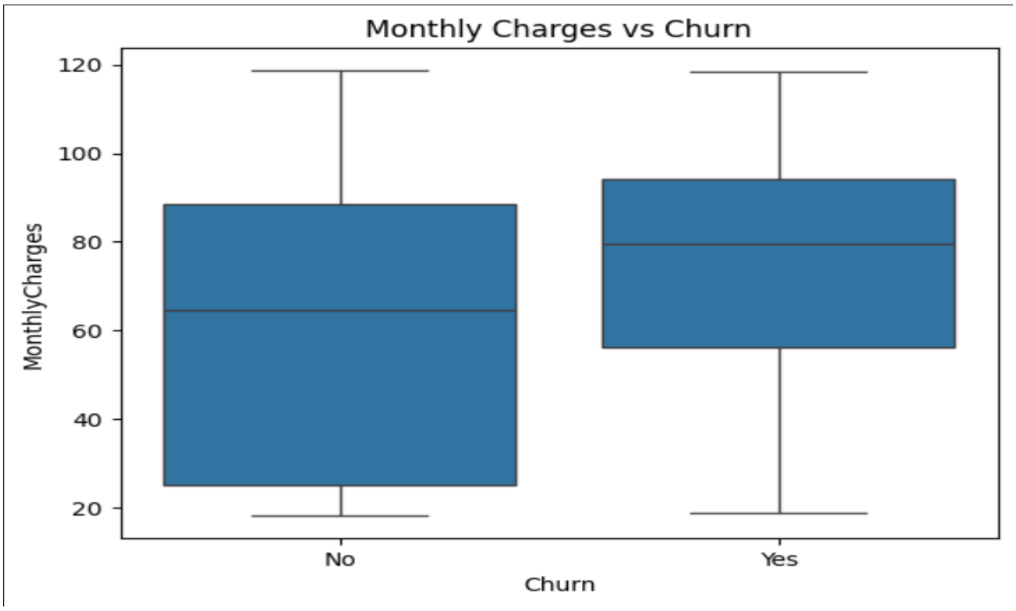
Business Interpretation

Customers may perceive high pricing as a burden and seek alternative providers offering similar services at lower costs.

Key Observation

Monthly charges positively influence churn behaviour.

Output Reference: Monthly Charges vs Churn Boxplot



3.5 INTERNET SERVICE ANALYSIS

The dataset includes different categories of internet services such as:

- DSL
- Fiber Optic
- No Internet Service

The analysis indicated that Fiber Optic customers experience higher churn rates than DSL customers.

This observation suggests that customers using premium internet services may have higher expectations regarding service quality and pricing.

Key Findings

Internet Service	Churn Trend
Fiber Optic	Highest
DSL	Moderate
No Internet Service	Lowest

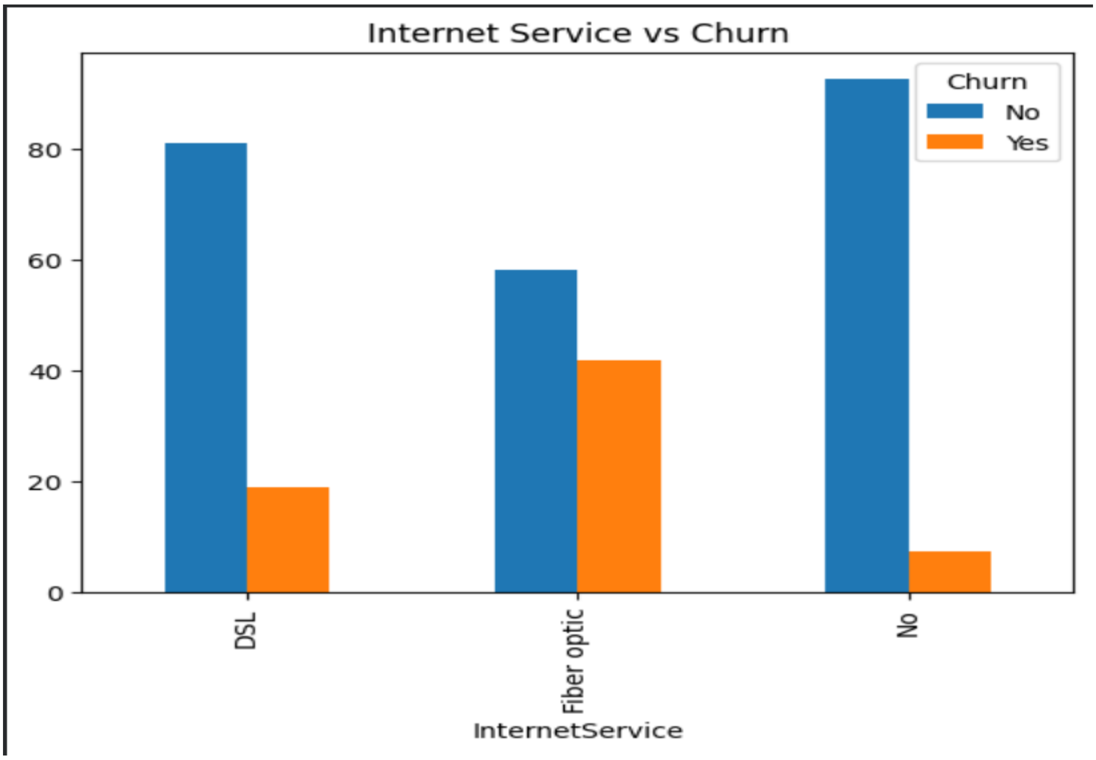
Business Interpretation

Organizations should investigate service quality, pricing strategies, and customer satisfaction levels among Fiber Optic users.

Key Observation

Internet service type significantly affects customer churn.

Output Reference: Internet Service vs Churn Bar Chart



3.6 CORRELATION MATRIX ANALYSIS

A correlation matrix was generated to identify relationships among variables.

Correlation values range from -1 to +1.

- Positive correlation indicates a direct relationship.
- Negative correlation indicates an inverse relationship.
- Values near zero indicate weak relationships.

The heatmap visualization provides a comprehensive overview of feature relationships.

Important Correlations

Tenure and Churn

A negative relationship was observed between tenure and churn.

Customers with longer tenure are less likely to leave.

Monthly Charges and Churn

A positive relationship was observed between monthly charges and churn.

Higher monthly charges increase churn likelihood.

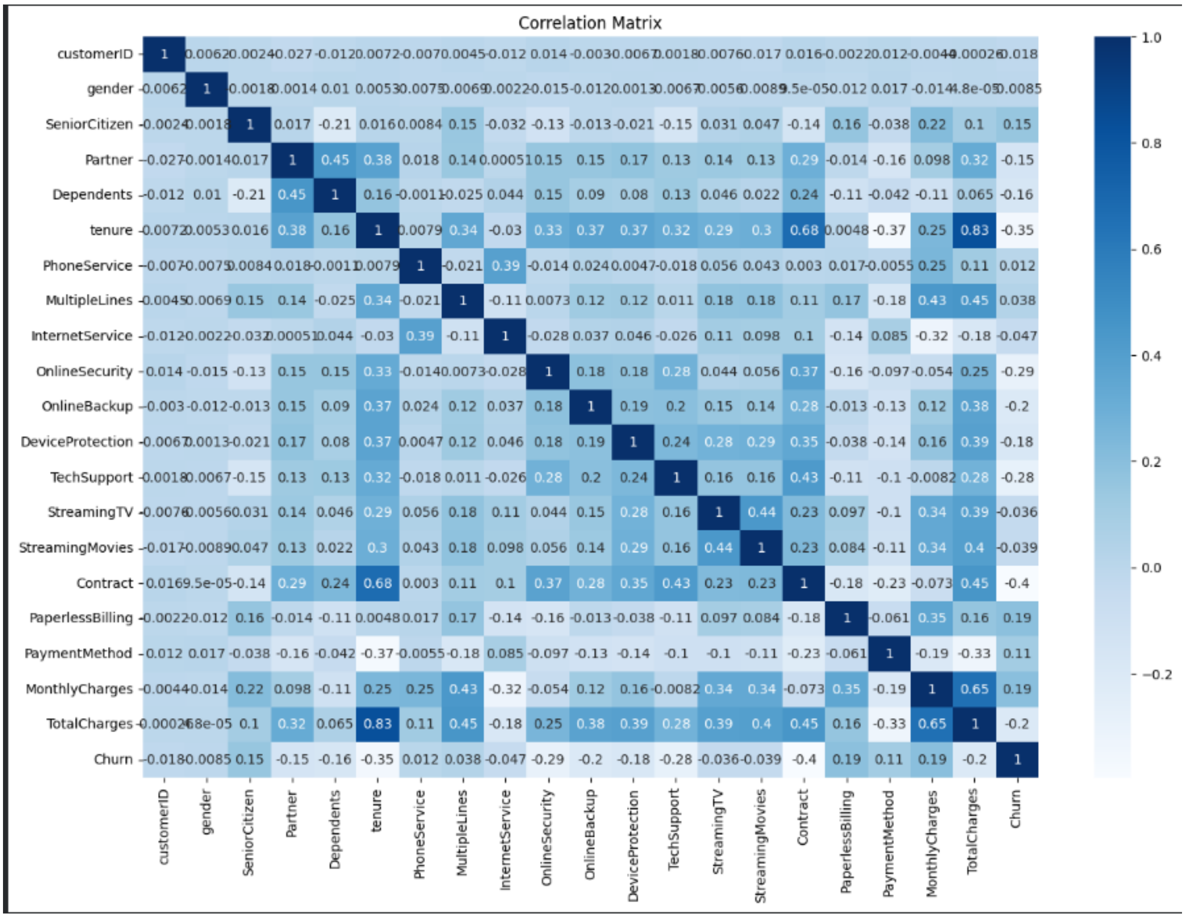
Contract and Churn

Contract duration demonstrated a strong relationship with customer retention.

Business Interpretation

Correlation analysis helps identify variables that significantly influence customer churn and improves feature selection for machine learning models.

Output Reference: Correlation Matrix Heatmap



3.7 KEY BUSINESS INSIGHTS

Based on the exploratory analysis, several important business insights were identified.

Insight 1: Contract Type is the Strongest Churn Factor

Customers with month-to-month contracts are more likely to churn than customers with long-term contracts.

Insight 2: New Customers are High-Risk Customers

Most churn occurs during the early stages of customer engagement.

Insight 3: Pricing Influences Customer Decisions

Customers paying higher monthly charges exhibit greater churn tendencies.

Insight 4: Fiber Optic Customers Require Attention

Fiber Optic users show higher churn rates and should be targeted through customer retention programs.

Insight 5: Customer Retention is More Effective than Customer Acquisition

Identifying high-risk customers early can significantly reduce revenue loss and improve long-term customer satisfaction.

MACHINE LEARNING MODELLING

4. MACHINE LEARNING MODELLING

Machine Learning plays a vital role in predictive analytics by enabling systems to learn patterns from historical data and make accurate predictions on unseen data. In this project, machine learning techniques were utilized to predict customer churn based on customer demographics, service information, billing details, and subscription behaviour.

After completing data preprocessing and exploratory data analysis, the prepared dataset was used for model development. Multiple classification algorithms were implemented and evaluated to identify the most suitable model for churn prediction. The performance of each model was assessed using standard evaluation metrics to ensure reliable prediction results.

The primary objective of this chapter is to develop predictive models capable of identifying customers who are likely to discontinue the company's services.

4.1 Logistic Regression

Logistic Regression is one of the most widely used classification algorithms in machine learning. It is particularly effective for binary classification problems where the target variable consists of two possible outcomes.

In this project, Logistic Regression was implemented as a baseline model for customer churn prediction. The algorithm estimates the probability of a customer belonging to either the churn or non-churn category based on input features.

The model was trained using the prepared training dataset and evaluated using the testing dataset. Logistic Regression provided a simple and interpretable approach for understanding customer churn behaviour.

Although Logistic Regression performs well for linear relationships, it may not fully capture complex interactions among customer attributes. Nevertheless, it serves as an important benchmark for comparing more advanced machine learning algorithms.

4.2 Decision Tree

Decision Tree is a supervised machine learning algorithm that uses a tree-like structure to make classification decisions. The algorithm divides the dataset into smaller subsets based on feature values and generates decision rules that lead to a final prediction.

The Decision Tree model was implemented to identify patterns associated with customer churn. Each internal node of the tree represents a decision based on a feature, while leaf nodes represent classification outcomes.

One of the major advantages of Decision Trees is their interpretability. The decision-making process can be easily visualized and understood by users.

The model successfully captured nonlinear relationships among customer attributes. However, Decision Trees may suffer from overfitting when the tree becomes excessively complex.

Despite this limitation, Decision Trees provide valuable insights into customer behaviour and contribute to churn prediction analysis.

4.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance and reduce overfitting.

The algorithm constructs numerous decision trees during training and generates predictions based on majority voting among individual trees. This approach increases model stability and improves generalization capability.

In this project, Random Forest was implemented as an advanced classification model for customer churn prediction. The model was trained using customer-related features and evaluated using the testing dataset.

Random Forest demonstrated strong predictive capability due to its ability to capture complex feature interactions and reduce variance. As a result, it achieved superior performance compared to the other classification models implemented in this study.

The algorithm also provides feature importance scores, enabling identification of the most influential variables affecting customer churn.

4.4 Model Evaluation

Model evaluation is essential for measuring the effectiveness of machine learning algorithms. After training the models, their performance was evaluated using the testing dataset.

Several evaluation metrics were considered to assess prediction quality.

- **Accuracy**

Accuracy measures the proportion of correctly classified instances relative to the total number of observations.

A higher accuracy value indicates better model performance.

- **Precision**

Precision measures the proportion of correctly predicted churn customers among all customers predicted as churners.

High precision indicates that the model produces fewer false positive predictions.

- **Recall**

Recall measures the proportion of actual churn customers correctly identified by the model.

High recall is particularly important for customer churn prediction because missing potential churners may result in customer loss.

- **F1 Score**

The F1 Score provides a balanced measure of precision and recall. It is useful when both false positives and false negatives are important.

The evaluation metrics enabled comparison of the implemented machine learning models and assisted in selecting the best-performing algorithm.

```
Logistic Regression Accuracy:
0.7853589196872779

Decision Tree Accuracy:
0.7249466950959488

Random Forest Accuracy:
0.7924662402274343

Random Forest Report
              precision    recall  f1-score   support

           0       0.83      0.90      0.86      1033
           1       0.64      0.49      0.56       374

    accuracy                           0.79      1407
   macro avg       0.74      0.70      0.71      1407
weighted avg       0.78      0.79      0.78      1407
```

4.5 Confusion Matrix Analysis

A confusion matrix was generated to provide a detailed evaluation of classification performance.

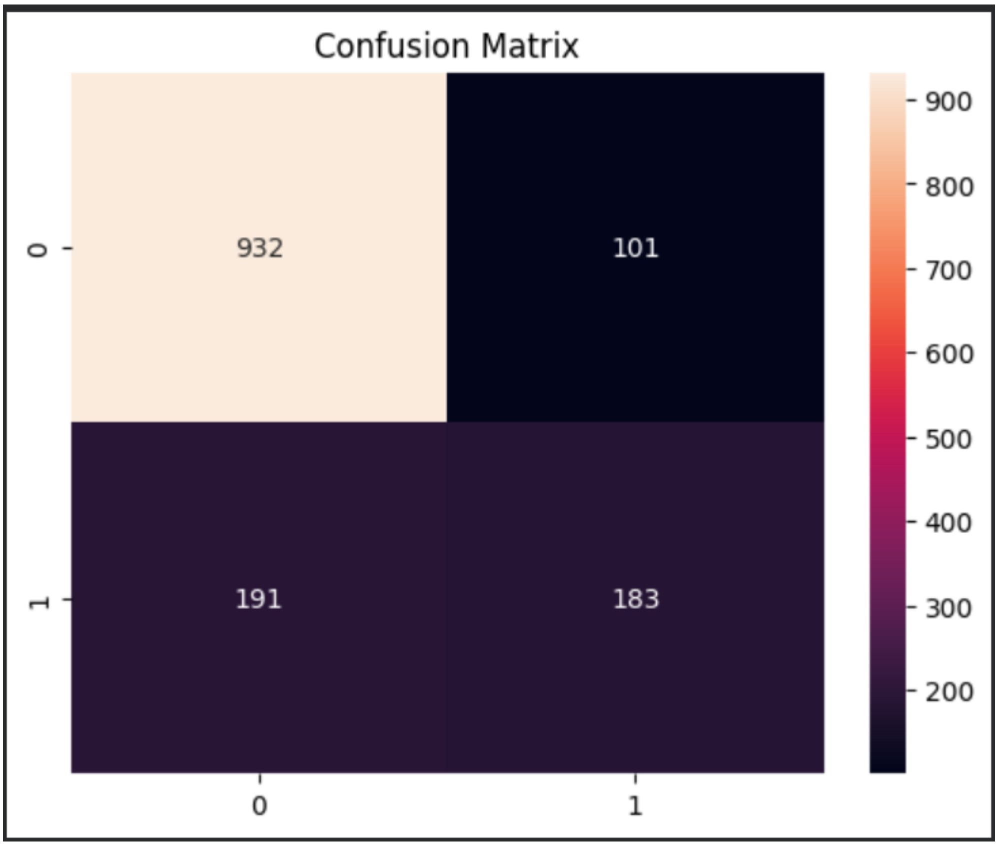
The confusion matrix consists of four components:

- True Positives
- True Negatives
- False Positives
- False Negatives

True Positives represent customers correctly predicted as churners, while True Negatives represent customers correctly classified as non-churners.

False Positives occur when non-churn customers are incorrectly predicted as churners, whereas False Negatives occur when actual churn customers are incorrectly classified as non-churners.

The confusion matrix provides deeper insight into model behaviour beyond overall accuracy and helps identify areas for improvement.



4.6 Feature Importance Analysis

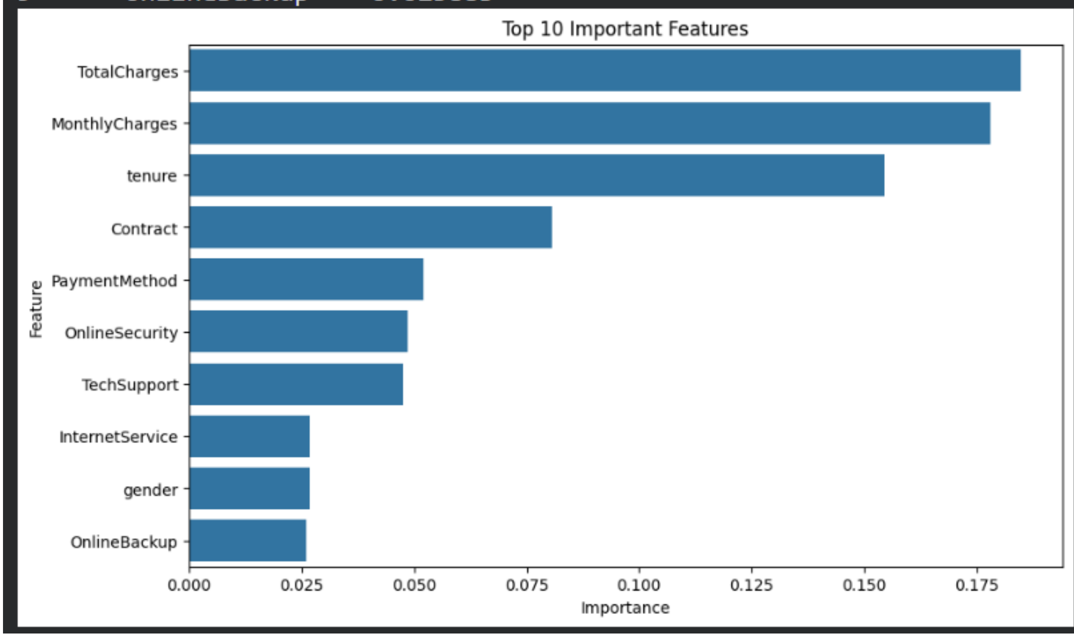
Feature importance analysis was performed using the Random Forest algorithm to identify variables that contribute most significantly to customer churn prediction.

The analysis revealed that several attributes play a major role in influencing churn behaviour. Features such as contract type, tenure, monthly charges, total charges, and internet service demonstrated strong predictive power.

Understanding feature importance is valuable because it allows organizations to focus on the factors that have the greatest impact on customer retention.

The results obtained from feature importance analysis support the findings observed during exploratory data analysis and further validate the significance of key churn-related variables.

```
Top 10 Important Features
            Feature  Importance
18     TotalCharges    0.184953
17   MonthlyCharges    0.178014
4            tenure    0.154510
14         Contract    0.080585
16    PaymentMethod    0.052031
8    OnlineSecurity    0.048509
11      TechSupport    0.047581
7   InternetService    0.026813
0            gender    0.026715
9      OnlineBackup    0.025883
```



4.7 Churn Risk Prediction System

A churn risk prediction system was developed using the best-performing machine learning model. The system predicts the probability of customer churn based on customer information and service attributes.

Each customer is assigned a churn probability score indicating the likelihood of discontinuing the service. Higher scores represent increased churn risk, while lower scores indicate greater customer loyalty.

The prediction system enables organizations to identify at-risk customers before they leave and take proactive measures to improve retention.

By automating the prediction process, businesses can make data-driven decisions and allocate retention resources more effectively.

```
   Risk Score (%)
0             0.0
1            10.0
2            75.0
3            21.0
4            24.0
```

4.8 Retention Recommendation Engine

In addition to churn prediction, a retention recommendation engine was developed to support customer retention strategies.

Based on churn probability scores, customers were categorized into different risk groups.

Customers with high churn probability were classified as High-Risk Customers and were recommended for immediate retention actions such as personalized offers, loyalty rewards, discounts, and dedicated customer support.

Customers with moderate churn probability were categorized as Medium-Risk Customers and targeted with promotional campaigns and service improvements.

Customers with low churn probability were considered stable customers and required only routine engagement activities.

The recommendation engine transforms predictive insights into actionable business strategies and enhances the practical value of the churn prediction system.

```
    Risk Score (%)          Recommendation
0              0.0      No Action Required
1             10.0      No Action Required
2             75.0          Offer Discount
3             21.0      No Action Required
4             24.0      No Action Required
5             55.0  Offer Loyalty Benefits
6             13.0      No Action Required
7             67.0  Offer Loyalty Benefits
8             22.0      No Action Required
9              2.0      No Action Required
10            50.0  Offer Loyalty Benefits
11            27.0      No Action Required
12             0.0      No Action Required
13            20.0      No Action Required
14             7.0      No Action Required
15            31.0      No Action Required
16           100.0          Offer Discount
17             9.0      No Action Required
18            33.0      No Action Required
19            25.0      No Action Required
```

4.9 Model Comparison

The performance of Logistic Regression, Decision Tree, and Random Forest models was compared using evaluation metrics obtained from the testing dataset.

The comparison demonstrated differences in predictive capability among the algorithms. Logistic Regression provided a simple baseline model, while Decision Tree captured nonlinear relationships within the data. Random Forest achieved the highest predictive performance due to its ensemble learning approach and ability to reduce overfitting.

Based on the evaluation results, Random Forest was selected as the final model for customer churn prediction.

The selected model demonstrated strong predictive accuracy and effectively identified customers at risk of churn, making it suitable for practical business applications.

The machine learning modelling process successfully transformed customer data into predictive insights and established a reliable foundation for customer retention decision-making.

	Model	Accuracy
0	Logistic Regression	0.785359
1	Decision Tree	0.724947
2	Random Forest	0.792466

RESULTS AND DISCUSSION

5. RESULTS AND DISCUSSION

The performance of the machine learning models was evaluated using the testing dataset to determine their effectiveness in predicting customer churn. The evaluation process focused on measuring the ability of each model to correctly classify customers as churners or non-churners. The results obtained from the implemented algorithms provide valuable insights into the predictive capability of the proposed churn prediction system.

The findings obtained from Exploratory Data Analysis and Machine Learning Modelling were carefully analysed to understand customer behaviour and identify effective retention strategies.

5.1 Model Evaluation Results

The implemented machine learning models, namely Logistic Regression, Decision Tree, and Random Forest, were evaluated using standard classification metrics. These metrics included accuracy, precision, recall, and F1-score.

The evaluation results indicated that all three models were capable of predicting customer churn; however, their performance varied depending on the complexity of the algorithm and its ability to capture relationships among customer attributes.

Logistic Regression provided a reliable baseline model and demonstrated satisfactory classification performance. Decision Tree improved interpretability by generating clear decision rules based on customer characteristics. Random Forest achieved superior predictive performance due to its ensemble learning approach and ability to reduce overfitting.

The evaluation process confirmed that machine learning techniques can effectively identify customers who are likely to discontinue services and assist organizations in implementing proactive retention measures.

5.2 Best Model Selection

The comparison of model performance revealed that Random Forest achieved the highest predictive accuracy among the implemented algorithms. The ensemble nature of Random Forest allowed it to capture complex patterns present within the customer data while maintaining strong generalization capability.

The superior performance of Random Forest can be attributed to its ability to combine predictions from multiple decision trees, thereby reducing variance and improving overall model stability. The model successfully identified important customer attributes contributing to churn and generated accurate predictions on unseen data.

Based on the evaluation metrics and comparative analysis, Random Forest was selected as the final model for customer churn prediction. The selected model provides a reliable framework for identifying high-risk customers and supporting retention-related business decisions.

5.3 Business Recommendations

The insights obtained from the analysis provide several practical recommendations for improving customer retention and reducing churn.The findings suggest that customers enrolled in month-to-month contracts are more likely to discontinue services. Organizations should therefore encourage customers to adopt long-term subscription plans through loyalty programs, promotional offers, and contract-based incentives.

The analysis also revealed that customers with shorter tenure exhibit higher churn tendencies. Businesses should focus on improving customer onboarding experiences and providing personalized support during the initial stages of subscription.

Higher monthly charges were associated with increased churn rates. Companies should regularly review pricing strategies and ensure that customers perceive sufficient value from the services provided. Flexible pricing plans and bundled service packages may help improve customer satisfaction.

Fiber Optic customers demonstrated relatively higher churn rates compared to other customer groups. Service providers should continuously monitor customer feedback, improve service quality, and address customer concerns to enhance retention among premium subscribers.

Implementing these recommendations can significantly improve customer loyalty and reduce revenue loss caused by customer attrition.

5.4 Discussion

The results obtained from this study demonstrate the effectiveness of machine learning techniques in predicting customer churn. The combination of data preprocessing, exploratory data analysis, feature selection, and predictive modelling enabled the identification of important factors influencing customer behaviour.

The findings indicate that customer churn is influenced by multiple factors rather than a single variable. Contract type, tenure, monthly charges, and internet service emerged as the most influential predictors of churn. These variables consistently appeared as important factors across both exploratory analysis and machine learning evaluation.

The developed churn prediction system can assist organizations in identifying high-risk customers before they discontinue services. Such early identification allows businesses to implement targeted retention strategies and improve overall customer satisfaction.

The study highlights the practical value of predictive analytics in customer relationship management and demonstrates how data-driven decision-making can contribute to organizational success.

CONCLUSION AND
FUTURE ENHANCEMENT

6.1 Conclusion

Customer churn prediction has become an important area of research and business application due to its direct impact on organizational profitability and customer retention. The primary objective of this project was to develop a machine learning-based system capable of predicting customer churn and supporting proactive retention strategies.

The Telco Customer Churn dataset was successfully analyzed using various data science techniques. Data preprocessing was performed to clean and prepare the dataset for analysis. Exploratory Data Analysis helped identify meaningful patterns and relationships among customer attributes.

The study revealed that contract type, tenure, monthly charges, and internet service significantly influence customer churn behavior. These findings provided valuable business insights and assisted in feature selection for predictive modelling.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated. Comparative analysis demonstrated that Random Forest achieved the best predictive performance and was selected as the final model for churn prediction.

A churn risk prediction system and retention recommendation engine were developed to transform predictive results into actionable business decisions. The proposed system enables organizations to identify high-risk customers and implement suitable retention strategies before customer attrition occurs.

Overall, the project successfully demonstrated the application of machine learning techniques in customer churn prediction and highlighted the importance of data-driven decision-making in customer relationship management.

6.2 Future Enhancement

Although the developed system provides reliable churn prediction capabilities, several enhancements can be incorporated in future work to further improve performance and functionality.

Future improvements may include the integration of real-time customer data for continuous churn monitoring and prediction. Deep learning algorithms such as Artificial Neural Networks and Long Short-Term Memory networks can be explored to improve predictive accuracy.

The system can also be integrated with Customer Relationship Management (CRM) platforms to automate retention campaigns and customer engagement activities. Cloud-based deployment can improve scalability and accessibility for large organizations.

Interactive dashboards and visualization tools may be developed to provide business managers with real-time insights regarding customer behavior and churn trends.

These enhancements have the potential to increase the effectiveness of the proposed system and extend its applicability across various business domains.

6.3 References

1. Han, J., Kamber, M., and Pei, J. Data Mining: Concepts and Techniques. Morgan Kaufmann Publishers.
2. Géron, A. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.
3. Witten, I. H., Frank, E., Hall, M. A., and Pal, C. J. Data Mining: Practical Machine Learning Tools and Techniques.
4. Scikit-Learn Documentation.
5. Pandas Documentation.
6. NumPy Documentation.
7. Matplotlib Documentation.
8. Seaborn Documentation.
9. Telco Customer Churn Dataset Documentation.
10. Research articles related to customer churn prediction and machine learning applications in telecommunications.